

Ameer Hamza Nasir

hamza6700@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, Bash, Asyncio, Software Development
- **Data Science & Analysis:** Pandas, NumPy, EDA, Feature Engineering, Statistical Analysis, Correlation Analysis, Data Cleaning, Data Visualization, Segmentation Modeling, Experiment Design, Matplotlib, Seaborn
- **Machine Learning & AI:** Scikit-learn, PyTorch, Supervised Learning, Random Forest, Regression, Classification, Clustering, ETL Pipelines, ML Frameworks, Model Evaluation (R^2 , MAE, RMSE, F1)
- **Deep Learning & Computer Vision:** Neural Networks, CNN, OpenCV, MNIST, Image Processing, Object Detection, PyTorch
- **MLOps & Deployment:** FastAPI, Streamlit, Flask, Docker, Kubernetes, GitHub Actions (CI/CD), Joblib, REST APIs, API Integration, Cloud Deployment
- **Databases & Tools:** SQL, SQLite, Data Validation, Git, Linux, Shell Scripting, Jupyter, Pytest

KEY PROJECTS

Intelligent Traffic Signal Optimizer | [View](#)

Pakistan

- Designed a real-time adaptive traffic signal system addressing Pakistan's fixed-timer problem (60–120s regardless of lane occupancy) using a Greedy scheduling algorithm to dynamically allocate green time based on live vehicle density.
- Trained a YOLOv5 vehicle detection model (PyTorch) on Pakistani road footage to count vehicles per lane in real-time via OpenCV frame capture.
- Implemented a priority queue (min-heap) in Python to rank lanes by congestion score, reducing average wait time by prioritizing high-density lanes over empty ones.
- Built a simulation dashboard in Python visualizing signal state transitions, per-lane queue lengths, and throughput metrics across a 4-way intersection.

Credit Card Fraud Detection System | [View](#)

- Orchestrated an end-to-end fraud detection pipeline using Python and Bash scripting on a highly imbalanced dataset (492 fraud / 284,807 transactions), applying SMOTE oversampling and stratified k-fold cross-validation.
- Benchmarked Logistic Regression, Random Forest, and XGBoost; selected XGBoost achieving $F1 = 0.91$, $AUC-ROC = 0.98$ with optimized decision threshold tuning.
- Performed full EDA: transaction distribution analysis, correlation heatmaps, and PCA visualization of feature clusters to identify fraud patterns.
- Deployed the model as a FastAPI REST endpoint on an AWS EC2 Linux server, containerized with Docker and automated with Bash scripts for reproducible cloud deployment.

Handwritten Digit Recognition | [View](#)

- Built and trained a Convolutional Neural Network (CNN) in PyTorch on the MNIST dataset (60,000 training / 10,000 test images) achieving 99.1% test accuracy.
- Designed architecture: 2 conv layers (ReLU + MaxPool) 'dropout' 2 fully connected layers; trained with Adam optimizer and cross-entropy loss over 10 epochs.
- Deployed as an interactive Streamlit web app with a canvas draw interface - users draw digits and get real-time CNN predictions with confidence scores.

Heart Disease Risk Predictor | [View](#)

- Executed full EDA on a Kaggle healthcare dataset: profiled missing data, generated correlation matrices, and produced visualization reports identifying top risk factors (age, cholesterol, max heart rate).
- Trained and compared Logistic Regression, Decision Tree, and Random Forest classifiers; selected best model by F1 score and ROC-AUC on held-out test set.
- Delivered findings as a reproducible Jupyter notebook with annotated statistical commentary, formatted for healthcare stakeholder review.

Student Exam Score Predictor | [View](#)

- Built full ETL pipeline processing 6,607 student records: ordinal encoding, binary encoding, feature selection ranking, and MinMax scaling.
- Trained Random Forest Regressor achieving $R^2 = 0.68$, $MAE = 2.03$; deployed as a Streamlit dashboard and production FastAPI endpoint containerized in Docker.
- Wrote pytest unit tests for ETL and model outputs integrated into GitHub Actions CI; visualized feature importance and predictions in interactive charts.

claudecode-free | [View](#)

- Architected an isolated Docker sandbox routing Claude Code API calls through a custom Node.js reverse-proxy to OpenRouter free-tier LLMs - eliminating Anthropic billing entirely.
- Engineered proxy.js: real-time Anthropic/OpenRouter format translation with request deduplication and SSE streaming support.
- Implemented GitHub Actions CI/CD to automate Docker image builds; repo gained organic stars from developers globally with zero promotion.

EDUCATION

Government College University, Faisalabad

B.S., Computer Science

• **Coursework:** C++, Calculus, ICT, Relational Databases & SQL

Oct 2025 - Present

Faisalabad

CERTIFICATIONS

• Python for Data Science 101: IBM / Cognitive Class

• SQL and Relational Databases: IBM / Cognitive Class

LANGUAGES

- English (Professional Working Proficiency)
- Urdu (Native)